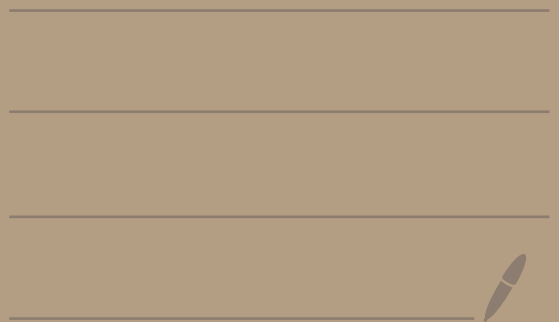


CSE 309

Week 8, Lecture 1

April 2, 2024.



April 2, 2024.

First 15 minutes : Quiz 3 on TMs.

Decidability

$$A_{DFA} = \{ \langle B, w \rangle : B \text{ is a DFA that accepts } w \}$$

Theorem: " A_{DFA} is decidable."

Proof: informal: Let T be a TM that decides A_{DFA} .

Now whenever some string x is passed to T , T will do following.

1. Decode B from x and simulate it.
2. Parse w from x to ①
3. If B accepts w , then accept, reject otherwise.

Theorem: " A_{NFA} is decidable."

$$A_{NFA} = \{ \langle B, w \rangle \mid B \text{ is an NFA that accepts } w \}$$

$$A_{\text{REX}} = \{ \langle R, w \rangle \mid R \text{ is a r.e. s.t. } w \text{ can be generated by } R \}$$

$$E_{\text{DFA}} = \{ \langle A \rangle \mid A \text{ is a DFA and } L(A) = \emptyset \}$$

$$EQ_{\text{DFA}} = \{ \langle A, B \rangle \mid A, B \text{ are DFA's s.t. } L(A) = L(B) \}$$